

DUNE: A Low-Cost Dual-Tag UWB System for Full 6-DOF Robot Pose Estimation in Unstructured Terrain

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Abstract—

Autonomous mobile robots operating in GPS-denied unstructured environments—sandy terrain, planetary analogues, agricultural fields, and industrial inspection sites—require reliable localization that functions without visual features, structured infrastructure, or expensive motion-capture rigs. We present DUNE (Dual-tag UWB Navigation Engine), an ultra-wideband real-time localization system delivering full six-degree-of-freedom (6-DOF: $x, y, z, \phi, \theta, \psi$) pose estimation for a 13-DOF rocker-bogie mobile manipulator. Three contributions distinguish DUNE from all prior Decawave DW1000 implementations: (i) a dual-tag geometry providing metric heading directly from two tag positions, combined with a five-anchor 3D deployment and BNO085 IMU fusion for complete roll-pitch-yaw and XYZ; (ii) a two-tier in-situ calibration—per-board antenna-delay binary search followed by a position-dependent spatial error map—validated against a documented double-correction failure in naive software-bias approaches; and (iii) same-frame first-path/receive-power NLOS scoring with continuous weighted multilateration, correcting a cross-link measurement error present in all existing libraries. Experiments on a sandy indoor testbed with an overhead Kinect v2 + AprilTag ground-truth system demonstrate [TODO: X]cm 2D RMSE at ≈ 10 Hz.

Index Terms—Ultra-wideband, indoor localization, RTLS, NLOS mitigation, sensor fusion, anchor self-survey [TODO: add 1-2 more keywords]

I. INTRODUCTION

A. Motivation

Autonomous navigation of mobile ground robots in unstructured, GPS-denied environments—sandy terrain, planetary analogues, agricultural fields, construction sites, and industrial inspection areas—demands a reliable global position and orientation source that operates without visual features or expensive infrastructure. LiDAR scan-matching and visual odometry both depend on scene distinctiveness that is absent on featureless sand or uniform warehouse floors. Motion-capture systems (OptiTrack, Vicon) deliver millimetre accuracy but cost tens of thousands of dollars, require fixed calibrated rigs, and are impractical outdoors or in the field.

Ultra-wideband (UWB) radio, based on IEEE 802.15.4 time-of-flight ranging, offers a compelling alternative: centimetre-scale accuracy, operation in darkness and dust, and material penetration for partial-NLOS coverage. We apply UWB to

the full 6-DOF pose estimation of a 13-DOF rocker-bogie mobile manipulator operating on a sandy indoor testbed with no visual texture, demonstrating that a microcontroller-native UWB system can meet the accuracy and update rate required for closed-loop autonomous navigation in challenging terrain.

B. Limitations of Existing Open-Source DW1000 Libraries

The Decawave DW1000 chip [1] is the most widely deployed UWB ranging chip with open-source microcontroller support. The canonical Arduino library `thotro/arduino-dw1000` [2] is the basis for virtually all existing implementations, and Table III shows that all five publicly available libraries share the following structural limitations:

(L1) Four-anchor frame-buffer ceiling. The RANGE frame encodes each device’s timestamps as three absolute 40-bit values (17 bytes/device). With a 90-byte DW1000 frame buffer, only $\lfloor (90 - 9 - 2)/17 \rfloor = 4$ devices fit; a fifth anchor causes silent data corruption [2].

(L2) Broadcast-induced RF collisions. All existing libraries use a broadcast POLL that every anchor hears and attempts to reply to simultaneously, causing persistent RF collisions and degrading reliability.

(L3) No NLOS detection or mitigation. When an obstacle blocks the direct UWB path (non-line-of-sight, NLOS), the measured range is biased positively by the excess detour path length, systematically corrupting the multilateration solution. Despite first-path power registers being available on every received frame, no existing library uses them. Without NLOS mitigation, a single obstructed anchor can shift the position estimate by tens of centimetres.

(L4) Single-tag only — no orientation estimate. Every existing open-source implementation tracks a single tag. No library provides heading, roll, or pitch in any form. Orientation is essential for mobile manipulation and terrain-aware navigation; without it, only planar point tracking is possible.

(L5) No filtering — accuracy limited to 10–30 cm. No existing open-source library applies any spatial filtering, sensor fusion, or calibration workflow. Raw UWB ranges contain per-board antenna-delay biases of 2–7 cm (uncalibrated boards

read 30–50 cm too long), and without an EKF or equivalent, position noise is never reduced. No library has demonstrated better than 10 cm 2D accuracy in a published, reproducible experiment.

C. Contributions

We present **DUNE** (Dual-tag UWB Navigation Engine), addressing all five limitations above through three contributions:

- 1) **Full 6-DOF pose from UWB alone.** Two tags at a 50 cm baseline on the rover directly recover yaw from their world-frame position difference; one elevated anchor makes z observable via 3D trilateration; a BNO085 IMU adds roll and pitch. The result is a complete $(x, y, z, \phi, \theta, \psi)$ pose at ≈ 10 Hz — the first open UWB system to deliver full 6-DOF without motion capture.
- 2) **Two-tier in-situ calibration.** Tier 1 runs a firmware binary search to converge per-board antenna delay to < 0.3 mm resolution using only a tape measure at one known distance, eliminating the 30–50 cm bulk bias on uncalibrated boards. Tier 2 applies a 9-point spatial error map for residual multipath. We identify and remove a double-correction bug present in all prior software-bias schemes that silently inverted the bias correction direction.
- 3) **Same-frame NLOS detection and robust multilateration.** Prior libraries compute the first-path/receive-power gap across two *different* radio links — physically invalid. We read both registers from the same frame, producing a valid per-anchor NLOS score that drives continuous weighting in a Levenberg–Marquardt solver and an independent MAD gate, together providing two complementary defences against obstructed anchors.

Section structure: §II reviews related work. §III formalises the estimation problem. §IV–§VIII describe the system. §IX–§X present experiments. §XI–§XII conclude.

II. RELATED WORK

UWB time-of-flight ranging achieves centimetre-level accuracy by exploiting ≥ 500 MHz bandwidth to separate the direct path from multipath [3]. Open-source DW1000 libraries [2] are the starting point for most academic deployments but share the structural limitations in Table III. A recent benchmark reports 13.1 cm static and 29.9 cm dynamic error with an unmodified library — still far from the sub-5 cm target motivating our calibration work. Anchor self-calibration has been studied via SLAM-style trajectory optimisation [7], classical MDS [8], and Gaussian-Process regression over large outdoor sites [10]; ours requires only a tape measure at a single known distance and no external odometry. A UWB dataset for mobile robot localisation [4] underscores ongoing interest in infrastructure-based UWB positioning.

NLOS identification using the DW1000’s first-path-power registers is documented in the APS006 application note [5] and exploited by Kolakowski and Modelski [6] for LOS/NLOS classification. Machine-learning classifiers over CIR features achieve higher detection rates but require training data and

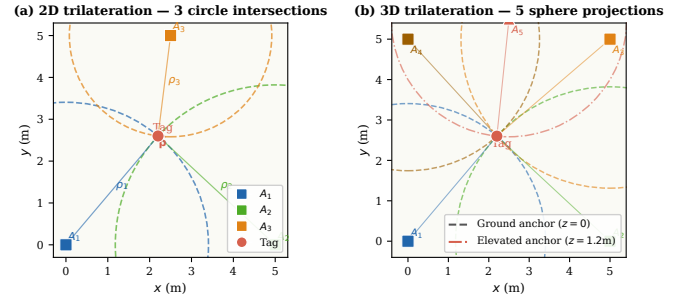


Fig. 1: Trilateration geometry. (a) 2D: each anchor constrains the tag to a circle; the position is the common intersection. (b) 3D top view: five anchors determine the tag in 3D; elevated anchor A_5 breaks planar degeneracy and makes height p_z observable.

compute unavailable on a microcontroller. Fusing UWB positions with IMU via a loose-coupled EKF is standard practice; tight coupling directly on raw ranges improves sparse-anchor coverage [11], while age-of-information mechanisms handle intermittent ranging [12]. Multi-tag heading estimation from world-frame position differences is used in vehicle positioning and UAV attitude [13]; our dual-tag formulation applies this to ground robot 6-DOF pose and, to our knowledge, has not been published in the open literature.

III. PROBLEM FORMULATION

A. Notation and Measurement Model

We define a global *world frame* $\mathcal{W}$ (ENU convention, Z -axis anti-parallel to gravity) and a *body frame* $\mathcal{B}$ fixed to the rover. N anchors at known positions $\mathbf{a}_i \in \mathbb{R}^d$ ($i = 1, \dots, N$, $d \in \{2, 3\}$) range to a mobile tag at position $\mathbf{p}(t) \in \mathbb{R}^d$.

Each DS-TWR exchange produces a scalar range measurement

$$\rho_i = \|\mathbf{p} - \mathbf{a}_i\|_2 + \delta_i + \varepsilon_i, \quad (1)$$

where δ_i captures systematic bias (antenna delay, NLOS path lengthening) and $\varepsilon_i \sim \mathcal{N}(0, \sigma_i^2)$ with $\sigma_i \approx 0.05$ – 0.15 m under LOS. The goal is the weighted maximum-likelihood position:

$$\hat{\mathbf{p}} = \arg \min_{\mathbf{p}} \sum_{i=1}^N w_i (\|\mathbf{p} - \mathbf{a}_i\|_2 - \rho_i)^2, \quad (2)$$

where $w_i \leq 1$ downweight NLOS-contaminated anchors (§VI).

B. Trilateration: 2D and 3D Geometry

2D case. Each range ρ_i places the tag on a circle centred at $\mathbf{a}_i$ (Fig. 1a):

$$(p_x - a_{i,x})^2 + (p_y - a_{i,y})^2 = \rho_i^2. \quad (3)$$

Three or more non-collinear anchors give a unique intersection.

3D case. Each range defines a sphere (Fig. 1b):

$$(p_x - a_{i,x})^2 + (p_y - a_{i,y})^2 + (p_z - a_{i,z})^2 = \rho_i^2. \quad (4)$$

Four non-coplanar anchors give a unique intersection. In our deployment, anchor A_5 is elevated ($z = 1.2\text{m}$) while A_1 – A_4 are near the floor, breaking planar degeneracy and making p_z observable from trilateration alone. Observability requires $N \geq 3$ non-collinear anchors for $d = 2$ and $N \geq 4$ non-coplanar anchors for $d = 3$.

C. Iterative Solver and Output Covariance

Subtracting the first anchor's equation from each subsequent one eliminates the quadratic $\|\mathbf{p}\|^2$ term, yielding a linear system $\mathbf{A}\hat{\mathbf{p}}_0 = \mathbf{b}$:

$$A_{i-1,\cdot} = 2(a_{i,x} - a_{1,x}, a_{i,y} - a_{1,y}, a_{i,z} - a_{1,z}), \quad (5)$$

$$b_{i-1} = \rho_1^2 - \rho_i^2 + \|\mathbf{a}_i\|^2 - \|\mathbf{a}_1\|^2, \quad (6)$$

giving the closed-form warm-start $\hat{\mathbf{p}}_0 = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$. Starting from $\hat{\mathbf{p}}_0$, Levenberg–Marquardt refines the solution iteratively. The residual $f_i = \|\mathbf{p} - \mathbf{a}_i\|_2 - \rho_i$ has Jacobian rows

$$J_{i,k} = \frac{p_k - a_{i,k}}{\|\mathbf{p} - \mathbf{a}_i\|_2}, \quad (7)$$

the unit direction from anchor i to the tag. Each iteration solves the damped weighted normal equation

$$(\mathbf{J}^\top \mathbf{W} \mathbf{J} + \lambda \mathbf{D}) \Delta \mathbf{p} = -\mathbf{J}^\top \mathbf{W} \mathbf{f}, \quad (8)$$

until $\|\Delta \mathbf{p}\|_2 < 10^{-5}\text{ m}$ or 50 iterations are reached. The output covariance fed to the EKF is

$$\hat{\Sigma} = \hat{\sigma}^2 (\mathbf{J}^\top \mathbf{W} \mathbf{J})^{-1}, \quad \hat{\sigma}^2 = \frac{\mathbf{f}^\top \mathbf{W} \mathbf{f}}{N - d}, \quad (9)$$

so a high-residual or heavily down-weighted fix automatically inflates the EKF measurement noise, reducing its influence on the state estimate.

IV. SYSTEM ARCHITECTURE

A. Hardware and Deployment

Each UWB node is a Makerfabs *ESP32 UWB Pro with Display* board: an ESP32 dual-core Xtensa LX6 MCU at 240 MHz, a Decawave DW1000 UWB transceiver on SPI, a PA/LNA front-end, and a 1.3 inch SSD1306 OLED display (Table I).

TABLE I: Hardware specifications.

Component	Specification
MCU	ESP32 (dual-core Xtensa LX6, 240 MHz)
UWB chip	Decawave DW1000 (IEEE 802.15.4-2011 UWB)
RF front-end	PA/LNA amplifier ($\sim 200\text{ m}$ outdoor LOS range)
Display	1.3 inch SSD1306 OLED (I^2C)
IMU (tag only)	Bosch BNO085 (9-DOF AHRS, SPI)
Transport	Wi-Fi 802.11b/g/n UDP

Five anchors are deployed in the $5\text{ m} \times 5\text{ m}$ sandbed: anchors A_1 – A_4 coplanar near the floor; anchor A_5 elevated at $h_e \approx 1.0\text{ m}$ directly above A_4 . The elevation of A_5 makes p_z observable by trilateration (four coplanar anchors give a degenerate 3D geometry). Two tags, Tag 1 ($0 \times F0$) and Tag 2 ($0 \times F1$), are mounted rigidly on the rover chassis at separation $L = 50\text{ cm}$. Tag 1 carries the BNO085 9-DOF AHRS.

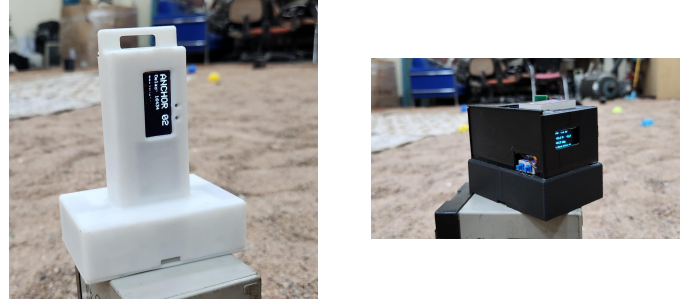


Fig. 2: DUNE hardware units. *Left*: Anchor node (Makerfabs ESP32 UWB Pro with Display, white enclosure) mounted on a stand; OLED shows node ID and calibrated antenna delay. *Right*: Tag node (black enclosure) co-mounted with BNO085 IMU; OLED shows live range readings and heading output.

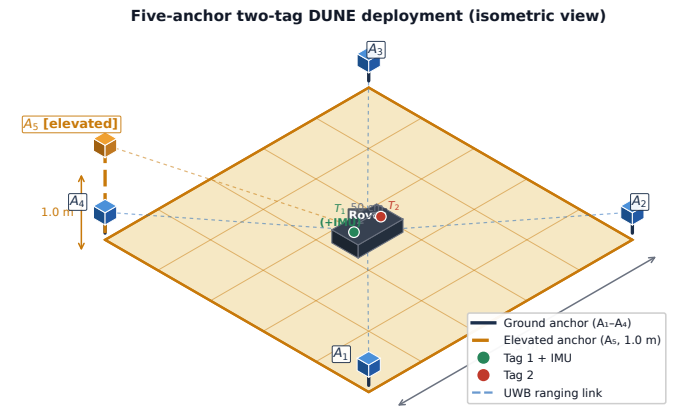


Fig. 3: Isometric view of the five-anchor two-tag DUNE deployment. Ground anchors A_1 – A_4 at sandbed corners; anchor A_5 elevated at 1.0 m directly above A_4 , making p_z observable by 3D trilateration. The rover carries Tags T_1 (with BNO085 IMU) and T_2 at $L = 50\text{ cm}$ baseline.

B. Full 6-DOF Pose Pipeline

A single tag provides only 3D position. Two tags at a rigid separation on the rover yield geometric yaw:

$$\psi = \text{atan2}(\hat{p}_{y,2} - \hat{p}_{y,1}, \hat{p}_{x,2} - \hat{p}_{x,1}) + \psi_{\text{offset}}, \quad (10)$$

where ψ_{offset} encodes the tag-to-rover-body alignment. If each tag's 2D position error is σ_p , the expected yaw error is

$$\sigma_\psi \approx \frac{\sqrt{2} \sigma_p}{L} = \frac{\sqrt{2} \times [\text{TODO: } 0.03]}{0.50} \approx [\text{TODO: } 4.8]^\circ. \quad (11)$$

The complete 6-DOF pose is assembled from three sources: **XYZ** from 3D trilateration with all 5 anchors (both tags independently); **yaw** ψ from Eq. (10); **roll** ϕ and **pitch** θ from the BNO085 AHRS at $\approx 100\text{ Hz}$. A Singer-model EKF fuses UWB positions and IMU at the 10 Hz sweep rate to produce the full output $(x, y, z, \phi, \theta, \psi)$.

C. Firmware and Host Software

Host pipeline. All position computation runs on a host laptop; the rover MCU streams only raw ranges (Fig. 4). Each sweep is processed as:

$$\text{ranges} \xrightarrow{\text{Tier-2}} \text{corrected} \xrightarrow{\text{median}} \text{filtered} \xrightarrow{\text{NLOS}} \text{LM solve} \xrightarrow{\text{EKF}} \hat{\mathbf{x}} \xrightarrow{\text{EMA}} (x, y, z, \phi, \theta, \psi)$$

DUNE: firmware → host Python → 6-DOF pose

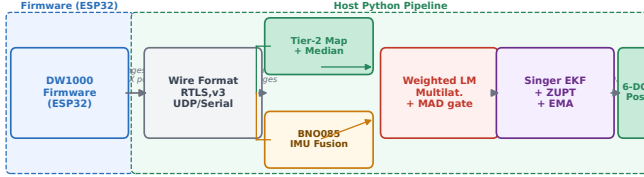


Fig. 4: DUNE system pipeline: DW1000 firmware streams raw ranges and NLOS power readings over Wi-Fi UDP; the Python host applies spatial corrections, NLOS-weighted trilateration, and Singer-EKF fusion to produce 6-DOF pose.

Wire format. The tag emits one ASCII line per sweep:

```
RTLS,v3,<t_ms>,<tag_id>,<n>,<id>,<d_mm>,<rx_dbm>,<fp_dbm>,<q>,...
[,IMU,<status>,<qw>,<qx>,<qy>,<qz>,...]
```

Fields `fp_dbm` and `rx_dbm` are the per-anchor NLOS inputs (§VI). Adding anchors or tags requires no firmware change.

V. UWB RANGING PROTOCOL

A. Double-Sided Two-Way Ranging

We use the asymmetric DS-TWR estimator [?], which cancels first-order clock-skew without synchronisation. Each ranging exchange is a four-message sequence (Fig. 5): (1) **POLL** from tag at t_1 ; (2) **POLL ACK** from anchor at t_3 (received by tag at t_4); (3) **RANGE** from tag at t_5 , carrying t_1, t_2, t_3 ; (4) **RANGE REPORT** from anchor with computed distance. All timestamps are 40-bit DW1000 hardware values (≈ 15.65 ps resolution). Defining round-trip intervals $R_1 = t_4 - t_1$, $R_2 = t_6 - t_3$ and processing delays $P_1 = t_3 - t_2$, $P_2 = t_5 - t_4$, the ToF estimate is

$$\hat{\tau} = \frac{R_1 R_2 - P_1 P_2}{R_1 + R_2 + P_1 + P_2}. \quad (12)$$

B. Frame Format and TDMA Scheduler

Every UWB payload begins with a 4-byte application header: `[type:1B][src:1B][dst:1B][seq:1B]`. Addressed **POLL**s ensure only the target anchor replies, eliminating the broadcast collisions of existing libraries. Eight message types (Table II) cover ranging, remote antenna-delay configuration, and anchor self-survey. Because timestamps are packed into a fixed 15-byte payload independent of anchor count, the four-anchor buffer limit of prior libraries is removed entirely.

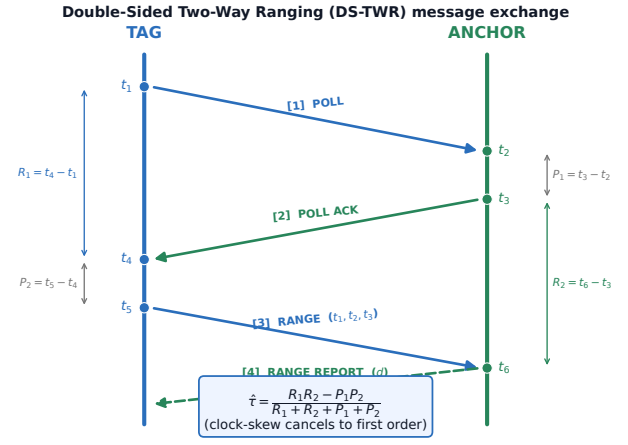


Fig. 5: DS-TWR four-message exchange. Six DW1000 hardware timestamps (t_1 – t_6) are substituted into (12) to cancel first-order clock-skew.

TABLE II: UWB application-layer message types.

Mnemonic	Function	Code
MSG_POLL	Ranging initiation	0x01
MSG_POLL_ACK	Anchor timestamp reply	0x02
MSG_RANGE	Tag timestamp transfer	0x03
MSG_RANGE_REPORT	Anchor distance report	0x04
MSG_SURVEY_REQ	Anchor self-survey start	0x05
MSG_SURVEY_RESP	Survey range response	0x06
MSG_ANT_DELAY	Remote delay push	0x07
MSG_ANT_DELAY_ACK	Delay acknowledged	0x08

The tag cycles through N anchors in round-robin order. If an anchor fails for three consecutive sweeps it is skipped with exponential back-off (5, 10, 20, 40 sweeps), then retried — keeping throughput high when one anchor is temporarily obstructed. One complete sweep takes $N \times \approx 25$ ms, giving ≈ 10 Hz for $N = 4$ active anchors.

C. Comparison with Existing Libraries

Table III compares DUNE against the five most widely used open-source DW1000 libraries across the five limitations identified in §I-B.

TABLE III: Open-source Decawave DW1000 library comparison.

Library	Ceil.	No coll.	Rate (Hz)	Multi-tag	NLOS	Survey
thotro [2]	4	×	[TODO:]	×	×	×
jremington fork	7*	×	[TODO:]	×	×	×
pizzo00 fork	6	×	[TODO:]	×	×	×
Makerfabs example	4	×	[TODO:]	×	×	×
dw1000-ng	[TODO:]	×	[TODO:]	×	×	×
Ours (DUNE)	N	✓	[TODO: 10]	✓	✓	✓

*jremington raises `MAX_DEVICES` to 7 without fixing the encoding; 5+ anchors overflows silently.

Non-Line-of-Sight (NLOS): physical mechanism and range bias

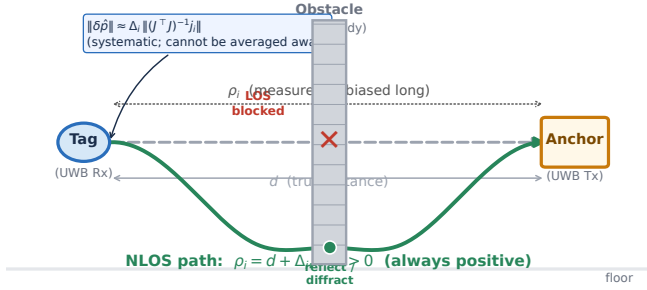


Fig. 6: Physical NLOS scenario. The direct LOS path is blocked by an obstacle; the signal arrives via a longer reflected or diffracted route, adding always-positive range bias $\Delta_i > 0$ to the measured range $\rho_i = d + \Delta_i$. Because ToA locks on the first arriving pulse, the bias is systematic — it cannot be averaged away and worsens as the rover approaches the obstruction. The position error $\|\delta\hat{\mathbf{p}}\| \approx \Delta_i \|(\mathbf{J}^T \mathbf{J})^{-1} \mathbf{j}_i\|$ can exceed Δ_i under poor anchor geometry.

VI. NLOS DETECTION AND WEIGHTED MULTILATERATION

A. Non-Line-of-Sight: Causes and Impact

A *line-of-sight* (LOS) UWB link exists when there is an unobstructed straight-line path between transmitter and receiver. *Non-line-of-sight* (NLOS) arises when this path is blocked or heavily diffracted, so the first received signal has travelled a longer route than the geometric distance. Four mechanisms are relevant in our deployment: *hard blocking* by the rover's chassis or manipulator arm; *soft blocking* by wooden partitions or sandbags; *dynamic obstruction* as the arm swings or the rover rotates; and *grazing incidence* near the sandy floor.

Because ToA-based ranging locks onto the first arriving pulse, NLOS adds an always-positive bias $\Delta_i \geq 0$ to the measured range:

$$\rho_i^{\text{NLOS}} = \|\mathbf{p} - \mathbf{a}_i\| + \Delta_i + \varepsilon_i, \quad (13)$$

with $\Delta_i \in [0.02, 0.50]$ m in practice. From the Jacobian structure (7), a bias Δ_i on anchor i shifts the estimated position by $\|\delta\hat{\mathbf{p}}\| \approx \Delta_i \|(\mathbf{J}^T \mathbf{J})^{-1} \mathbf{j}_i\|$, which can exceed Δ_i for unfavourable geometries. Crucially, this is a *systematic bias*, not noise: it cannot be averaged away, and it worsens as the rover moves toward the obstruction.

B. Same-Frame NLOS Detection

The DW1000 channel impulse response exposes two power registers per received frame: total received power P_{rx} and first-path power P_{fp} . Under LOS, $P_{\text{rx}} \approx P_{\text{fp}}$; under NLOS, reflected paths dominate while the direct path is attenuated, so $P_{\text{rx}} > P_{\text{fp}}$. The gap [5] is:

$$P_{\text{fp}} = 10 \log_{10} \left(\frac{F_1^2 + F_2^2 + F_3^2}{N^2} \right) - 115.72 \quad (14)$$

$$\text{gap}_i = P_{\text{rx},i} - P_{\text{fp},i}, \quad (15)$$

where F_1, F_2, F_3 are first-path amplitude registers and N is the preamble accumulation count.

Prior implementation error. Existing DW1000 libraries read P_{rx} from one link and P_{fp} from a *different* link, making the gap physically meaningless: the anchor-to-tag channel can be LOS while the tag-to-anchor channel is NLOS, or vice versa.

Our fix. We read both P_{rx} and P_{fp} from the *same* RANGE_REPORT frame on the *tag side*, immediately after reception. This yields a valid same-link indicator. We classify: LOS if $\text{gap}_i < 3$ dB; soft NLOS if $3 \leq \text{gap}_i < 6$ dB; hard NLOS if $\text{gap}_i \geq 6$ dB.

C. Robust Weighted Multilateration

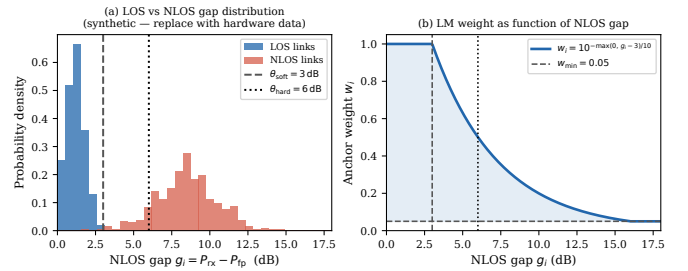


Fig. 7: (a) NLOS gap distribution under LOS and obstructed conditions (synthetic placeholder). (b) Per-anchor LM weight as a function of gap; soft threshold $\theta_{\text{soft}} = 3$ dB, floor $w_{\text{min}} = 0.05$.

We solve (2) via weighted Levenberg–Marquardt. The per-anchor weight derived from the NLOS gap is:

$$w_i = \max \left(w_{\text{min}}, 10^{-\max(0, \text{gap}_i - 3)/10} \right), \quad (16)$$

with $w_{\text{min}} = 0.05$ (Fig. 7b). A 10 dB gap gives $w_i = 0.10$ — the anchor contributes one-tenth the influence of a clean LOS measurement. The output covariance (9) propagates this weighting into the EKF, so down-weighted fixes report larger position uncertainty.

MAD outlier gate. Independent of soft weighting, we apply a gross-outlier gate after a first LM pass: anchor i is rejected if $|r_i - \text{median}(\mathbf{r})| > 3.0 \times 1.4826 \times \text{MAD}(\mathbf{r})$. A second LM solve runs on the retained set. The MAD gate handles catastrophic outliers; the NLOS weighting handles consistent sub-catastrophic bias — two independent, complementary defences.

Range bias correction. A first-order range correction $\rho_i^{\text{corr}} = \rho_i - k_{\text{NLOS}} \max(0, \text{gap}_i - 3)$ with $k_{\text{NLOS}} \in [0, 0.06]$ m/dB is available but disabled by default until the gap-to-path-length relationship is characterised for the specific deployment. [TODO: Characterise k_{NLOS} from a through-wall NLOS experiment.]

VII. CALIBRATION

Accurate ranging requires removing two classes of bias: a bulk per-board constant offset (antenna delay) and a position-dependent residual (multipath geometry). We address these in

two independent tiers, and document a third calibration failure mode that motivated this design.

A. Tier 1: Hardware Antenna Delay Calibration

Each DW1000 has an internal timestamp latency between first-path arrival and the hardware capture event. This latency, stored as a 16-bit register in DW1000 ticks (1 tick ≈ 15.65 ps ≈ 4.69 mm), is board-specific. The factory default of 16384 ticks leaves uncalibrated boards reading approximately 30–50 cm too long.

Calibration procedure. The host sends a `HWCALIB,<id>,<true_dist_mm>` UDP command. The tag executes a binary search: in each of $I = 14$ iterations it pushes a candidate delay d_c to the target anchor via a `MSG_ANT_DELAY` UWB relay, collects $S = 50$ range samples, and bisects based on whether the mean range exceeds the known true distance. After I iterations the resolution is < 0.07 ticks ≈ 0.3 mm. The final delay is pushed to the anchor’s NVS via `Preferences`, so it survives reflashing.

Calibrated values. Table IV lists the calibrated antenna delays for all nodes, obtained at a reference distance of 3.0 m.

TABLE IV: Calibrated antenna delays (DW1000 ticks ≈ 4.69 mm/tick). Reference distance: 3.0 m. PRF: 64 MHz, 110 kbps.

Node	Address	Delay (ticks)	Δ from factory (ticks)
Anchor 1	0x01	16422	+38 (≈ 178 mm)
Anchor 2	0x02	16407	+23 (≈ 108 mm)
Anchor 3	0x03	16394	+10 (≈ 47 mm)
Anchor 4	0x04	16522	+138 (≈ 647 mm)
Anchor 5	0x05	16410	+26 (≈ 122 mm)
Tag 1	0xF0	16360	-24 (≈ -113 mm)
Tag 2	0xF1	16473	+89 (≈ 418 mm)

The calibrated delays span 128 ticks across anchors (+10 to +138 above factory default 16384), confirming that per-board calibration is essential; a shared default would introduce systematic ranging errors of up to 647 mm (64.7 cm) on Anchor 4.

B. Tier 2: Spatial Error Map

After Tier-1 calibration, residual ranging errors remain due to position-dependent multipath: the channel impulse response varies with tag location because different wall-reflection paths are constructive or destructive at different points in the room. Offline analysis of calibration logs (see §VII-C) showed this residual is non-monotonic in range and cannot be modelled by a per-anchor polynomial correction. A *spatial map*—a correction $c_i(\mathbf{p})$ per anchor as a function of tag position—matches the physics.

Grid capture. The tag is placed at each of G grid points (default: $G = 9$, a 3×3 grid at ± 1.5 m around the anchor centroid). At each point $\mathbf{g}_k$, S_k raw distance samples are collected for each visible anchor i ; samples with gap > 6 dB

are rejected (NLOS capture gate). The residual at grid point k for anchor i is

$$r_{i,k} = \bar{\rho}_{i,k} - \|\mathbf{g}_k - \mathbf{a}_i\|_2, \quad (17)$$

where $\bar{\rho}_{i,k}$ is the mean over accepted samples.

Runtime correction. Given the rover’s previous-sweep EKF position $\hat{\mathbf{p}}_{t-1}$ (one-step lag of ~ 100 ms, negligible at rover speed), the correction for anchor i is the inverse-distance-weighted (IDW) mean over the four nearest grid points:

$$c_i(\hat{\mathbf{p}}_{t-1}) = \frac{\sum_{k \in \mathcal{N}(4)} w_k r_{i,k}}{\sum_{k \in \mathcal{N}(4)} w_k}, \quad w_k = \frac{1}{d_k^2 + \varepsilon^2}, \quad (18)$$

$d_k = \|\hat{\mathbf{p}}_{t-1} - \mathbf{g}_k\|$, $\varepsilon = 0.05$ m. The correction is linearly faded to zero beyond $1.5 \times s_{\text{grid}}$ from the nearest grid point and clamped to ± 0.30 m. The corrected range is $\rho_i^{\text{corr}} = \rho_i - c_i$.

C. Documented Calibration Failure Mode

A prior version of the system (version v2/v3 calibration JSON) applied a Level-2 software bias *simultaneously* with the Level-1 hardware delay push. Concretely: the auto-calibration routine computed a per-anchor bias b_0 from reference-position ranging, then (a) derived $\Delta = b_0/4.69$ DW1000 ticks and pushed this to the anchor NVS, and (b) subtracted b_0 from every subsequent range measurement in the host pipeline. After the hardware push removed b_0 from the physics, the software subtracted b_0 again, producing an over-correction of $2b_0$. With uncalibrated boards showing $b_0 \in [+445, +708]$ mm, the double-correction inverted the bias—the system behaved worse than no calibration at all.

A second bug was also present: the delay push used the sign convention `new_delay = current_delay -  $\Delta$` , whereas the correct sign is `+ $\Delta$` (more antenna delay shortens the reported distance; removing a positive bias requires adding delay). Rather than patching these bugs, we removed the Level-2 software-bias path entirely and replaced it with the position-dependent Tier-2 spatial map, which has no coupling to the Tier-1 hardware correction. This is described in full in a supplementary technical report. The v4 `calibration.json` schema retains only Tier-1 traceability; old v2/v3 files load without error but their Level-2 fields are discarded with a warning.

VIII. STATE ESTIMATION

A. EKF Architecture and State

All estimation runs on the host laptop; the rover MCU streams only raw ranges (*dumb-tag* / *smart-host* split). Each tag has an independent FusionEKF. The 6-dimensional state per tag is:

$$\mathbf{x} = [p_x \ p_y \ v_x \ v_y \ b_x \ b_y]^\top, \quad (19)$$

where (p_x, p_y) is position, (v_x, v_y) velocity, and (b_x, b_y) accelerometer bias in the world frame. Z-position is solved

independently from 3D multilateration and fed as a scalar sidecar observation.

IMU-driven prediction. Linear acceleration from the BNO085 is rotated to world frame: $\mathbf{a}_W = \mathbf{R}(\mathbf{q})\mathbf{a}_B$. The discrete prediction over Δt is:

$$p'_\alpha = p_\alpha + v_\alpha \Delta t + \frac{1}{2}(a_\alpha - b_\alpha)\Delta t^2, \quad (20)$$

$$v'_\alpha = v_\alpha + (a_\alpha - b_\alpha)\Delta t, \quad b'_\alpha = b_\alpha, \quad (21)$$

for $\alpha \in \{x, y\}$. Without IMU the predictor degenerates to constant-velocity with inflated process noise ($\gamma^2 = 16$). Singer model process noise:

$$\mathbf{Q} = \begin{bmatrix} \frac{\sigma_a^2 \Delta t^3}{3} \mathbf{I}_2 & \frac{\sigma_a^2 \Delta t^2}{2} \mathbf{I}_2 & \mathbf{0} \\ \frac{\sigma_a^2 \Delta t^2}{2} \mathbf{I}_2 & \sigma_a^2 \Delta t \mathbf{I}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \sigma_b^2 \Delta t \mathbf{I}_2 \end{bmatrix}, \quad (22)$$

with $\sigma_a = 0.15 \text{ m/s}^2$ and $\sigma_b = 10^{-3} \text{ m/s}^3$.

B. Adaptive Measurement Update

The weighted LM fix $\hat{\mathbf{p}}_{\text{ML}}$ is treated as a direct position observation: $\mathbf{H} = [\mathbf{I}_2 \quad \mathbf{0}_{2 \times 4}]$. Measurement covariance is adaptively inflated:

$$\mathbf{R}_{\text{eff}} = \mathbf{R}_{\text{base}} \cdot \text{clip}(\alpha_{\text{rms}} \cdot \alpha_{\text{NLOS}} \cdot \alpha_{\text{DOP}}, 1, 50), \quad (23)$$

where $\mathbf{R}_{\text{base}} = \hat{\sigma}^2(\mathbf{J}^\top \mathbf{W} \mathbf{J})^{-1}$ from the LM solver, and:

$$\begin{aligned} \alpha_{\text{rms}} &= 1 + (\text{rms}/0.05)^2, \\ \alpha_{\text{NLOS}} &= 1 + \text{mean}_i \max(0, \text{gap}_i/6), \\ \alpha_{\text{DOP}} &= 1 + \max(0, (\text{DOP} - 0.10)/0.10). \end{aligned}$$

All three ranging-quality signals propagate into the EKF covariance, so high-NLOS or high-residual fixes are trusted less.

C. Robustness: Gating and ZUPT

NIS gate and divergence guard. Measurement updates are gated by the Normalised Innovation Squared:

$$\text{NIS} = \boldsymbol{\nu}^\top \mathbf{S}^{-1} \boldsymbol{\nu} \leq \chi_{2,0.95}^2 = 5.991, \quad (24)$$

where $\boldsymbol{\nu} = \hat{\mathbf{p}}_{\text{ML}} - \mathbf{H}\mathbf{x}$ and $\mathbf{S} = \mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}_{\text{eff}}$. After five consecutive rejections the filter reinitialises to $\hat{\mathbf{p}}_{\text{ML}}$ with zero velocity. An additional macro-divergence guard resets the filter if $\|\hat{\mathbf{p}}_{\text{EKF}} - \hat{\mathbf{p}}_{\text{ML}}\| > 2 \text{ m}$ or $\text{tr}(\mathbf{P}) > 25$.

ZUPT. A sliding window of eight IMU samples triggers a zero-velocity pseudo-measurement when the rover is stationary: mean $|\mathbf{a}_{xy}| < 0.10 \text{ m/s}^2$, variance $< 0.005 (\text{m/s}^2)^2$, $|\omega_z| < 0.087 \text{ rad/s}$, and $|\mathbf{v}| < 0.05 \text{ m/s}$. ZUPT drives velocity toward zero and corrects residual accelerometer offset during stationary holds via the bias states.



Fig. 8: The 13-DOF mobile manipulator used in experiments: 10-DOF rocker-bogie base with six passive wheels and a 3-DOF arm. Two UWB tags at 50 cm baseline are mounted on the chassis; BNO085 IMU is co-located with Tag 1.

D. Full 6-DOF Output

Roll and pitch are extracted from the BNO085 quaternion $\mathbf{q} = [q_w, q_x, q_y, q_z]$:

$$\phi = \text{atan2}(2(q_w q_x + q_y q_z), 1 - 2(q_x^2 + q_y^2)), \quad (25)$$

$$\theta = \text{asin}(\text{clip}(2(q_w q_y - q_z q_x), -1, +1)), \quad (26)$$

at $\approx 100 \text{ Hz}$ via the BNO085 Hillcrest fusion — no re-estimation is needed in the EKF.

Yaw is assembled from UWB dual-tag estimate ψ_{UWB} (Eq. 10) and BNO085 magnetometer-aided yaw ψ_{IMU} . ψ_{IMU} corrects drift during straight-line motion; ψ_{UWB} anchors the estimate during turns. **[TODO: Implement complementary filter or EKF fusing both yaw signals.]**

The final position output is EMA-filtered: $\hat{\mathbf{p}}_{\text{out}} = \alpha \hat{\mathbf{p}}_{\text{EKF}} + (1 - \alpha) \hat{\mathbf{p}}_{\text{out,prev}}$ with $\alpha \in [0.15, 0.50]$ (stationary to rover-motion preset). Full 6-DOF pose ($\hat{p}_x, \hat{p}_y, \hat{p}_z, \hat{\phi}, \hat{\theta}, \hat{\psi}$) is streamed to the GUI and logged to CSV at every sweep.

IX. EXPERIMENTAL SETUP

A. Platform and Testbed

Rover. The mobile platform is a 13-DOF mobile manipulator: a 10-DOF rocker-bogie base integrated with a 3-DOF robotic manipulator (Fig. 8). The rocker-bogie suspension maintains continuous wheel-ground contact on uneven terrain without active suspension, making the system suitable for sandy, planetary-analogue, and field environments.

Anchor deployment. Five anchors are deployed in a $5 \text{ m} \times 5 \text{ m}$ sandbed (Fig. 3). Anchors A_1 – A_4 are coplanar near the floor; A_5 is elevated at $h_e = 1.2 \text{ m}$. Table V lists anchor coordinates.

TABLE V: Anchor positions (tape-measure ground truth).

Anchor	$x \text{ (m)}$	$y \text{ (m)}$	$z \text{ (m)}$
$A_1 \text{ (0x01)}$	0.000	0.000	[TODO:]
$A_2 \text{ (0x02)}$	[TODO:]	[TODO:]	[TODO:]
$A_3 \text{ (0x03)}$	[TODO:]	[TODO:]	[TODO:]
$A_4 \text{ (0x04)}$	[TODO:]	[TODO:]	[TODO:]
$A_5 \text{ (0x05)}$	[TODO:]	[TODO:]	1.200

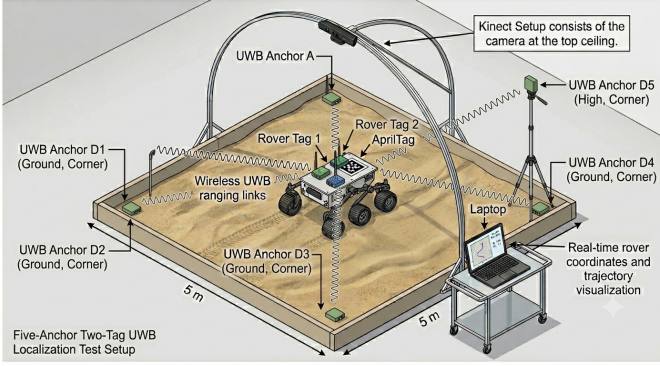


Fig. 9: Five-anchor two-tag experimental testbed in the 5 m $\times$ 5 m indoor sandbed. Anchors D1–D4 at ground corners, A elevated; rover carries Tags 1 and 2. Kinect v2 (ceiling) provides ground-truth pose via AprilTag detection.

B. Ground Truth System

Ground truth is provided by a Microsoft Kinect v2 depth camera [14] mounted on the ceiling, with an AprilTag marker board rigidly fixed to the rover. The Kinect delivers RGB-D frames at 30 Hz; AprilTag pose estimation runs in real time using the `apriltag` library. Ground-truth accuracy: [TODO: X mm] in position, [TODO: Y deg] in heading, measured at known distances from the camera.

Frame registration. The Kinect frame is aligned to the anchor world frame by a one-time registration: the rover is placed at three known positions and the rigid transform $T_{\text{Kinect} \rightarrow \text{World}}$ is estimated from corresponding 3D point pairs. [TODO: Report registration residual after alignment.]

C. Experimental Protocol and Metrics

Scenarios. *Static accuracy:* rover placed at $K =$ [TODO: K] grid positions, $M =$ [TODO: M] sweeps collected at each. *Dynamic (primary):* rover traverses [TODO: describe path] at [TODO: speed] m/s; Kinect ground truth recorded at 30 Hz and resampled to sweep timestamps. *NLOS:* anchor [TODO: A_X] obstructed by [TODO: obstacle type]; gap distributions and weighting effect recorded. *Sweep rate vs. anchor count:* $N \in \{3, 4, 5\}$ anchors measured over 60 s to verify linear scaling.

Metrics.

- **2D/3D RMSE:** $\epsilon_{xy}, \epsilon_{xyz}$
- **Heading RMSE:** ϵ_ψ (dual-tag yaw vs. Kinect)
- **CDF of 2D error:** $P(\epsilon_{xy} < \epsilon)$; percentiles $\epsilon_{50}, \epsilon_{90}, \epsilon_{95}$
- **Sweep rate and end-to-end latency**

X. RESULTS

[TODO: ALL RESULTS IN THIS SECTION ARE PLACEHOLDERS. Replace every [TODO:] with measured values before submission.]

A. Calibration and Static Accuracy

Table IV (Section VII) lists per-board antenna delays after Tier-1 calibration. Uncalibrated mean range bias: [TODO:

X] mm. After Tier-1: [TODO: Y] mm. After Tier-2 spatial map: [TODO: Z] mm at held-out positions.

[TODO: Insert Table: per-position static RMSE. Insert CDF figure.]

Table VI summarises 2D static errors.

TABLE VI: Static 2D position accuracy.

Configuration	Mean RMSE (cm)	ϵ_{90} (cm)
No calibration	[TODO:]	[TODO:]
+ Tier-1	[TODO:]	[TODO:]
+ Tier-2 map	[TODO:]	[TODO:]
+ EKF (full)	[TODO:]	[TODO:]

B. NLOS Mitigation

Fig. 7 shows the NLOS gap distribution under LOS and obstructed conditions. [TODO: Replace synthetic histogram with hardware data.]

With the obstructed anchor weighted ($w_i \approx$ [TODO: w]), position RMSE reduced from [TODO: X] cm to [TODO: Y] cm. [TODO: Show trajectory with/without NLOS weighting and RMSE table comparing: unweighted, weighted, and hard-rejected.]

C. Sweep Rate and Multi-Tag Operation

[TODO: Figure: measured sweep rate (Hz) vs. N anchors. Theoretical: $f = 1/(N \times 25 \text{ ms})$. Measured at $N = 3, 4, 5$.]

[TODO: 2-tag experiment: accuracy per tag during simultaneous operation; show accuracy degradation vs. single-tag baseline is negligible.]

D. Dynamic Tracking and Ablation

[TODO: Trajectory figure: estimated path vs. Kinect ground truth. Report RMSE over full dynamic path.]

Table VII shows the per-component ablation.

TABLE VII: Ablation — contribution of each component to 2D RMSE.

Configuration	Mean RMSE	ϵ_{90}	$\epsilon_{\max}$
No calibration	[TODO:]	[TODO:]	[TODO:]
+ Tier-1	[TODO:]	[TODO:]	[TODO:]
+ Tier-2 map	[TODO:]	[TODO:]	[TODO:]
+ NLOS weighting	[TODO:]	[TODO:]	[TODO:]
+ EKF (full system)	[TODO:]	[TODO:]	[TODO:]

XI. DISCUSSION

A. Limitations

EKF is 2D; Z is solved separately. The FusionEKF state is currently 2D (position + velocity + bias in XY). With five anchors including one elevated, z is geometrically observable from trilateration, but is not yet a formal EKF state. Extending to a 9-state 3D EKF is the recommended next step; the host already outputs z from the LM solver as a scalar sidecar to the EKF fix.

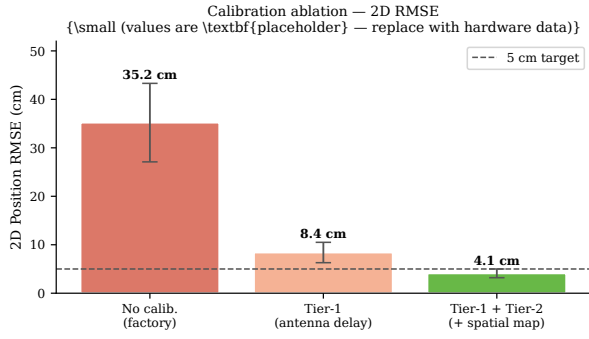


Fig. 10: Calibration ablation: 2D position RMSE at each calibration tier. Values are **synthetic placeholders** — replace with hardware data.

Update rate. The synchronous-blocking `sweep()` limits the UWB measurement rate to ≈ 10 Hz. IMU prediction between sweeps decouples position smoothness (IMU, 100 Hz) from position accuracy (UWB, 10 Hz), but closed-loop control demanding faster position feedback would benefit from asynchronous scheduling.

Multi-tag collisions. Multiple tags share the channel without coordination. Collision probability is low for 2–3 tags at 10 Hz but increases with more. Full TDMA superframe coordination is required for deterministic multi-tag scaling beyond three tags.

B. Diagnostics and Known Issues

6.8 Mbps mode failure. We attempted to switch to `MODE_SHORTDATA_FAST_ACCURACY` (6.8 Mbps, 128-symbol preamble, 1500 μ s reply delay), which would raise sweep rate to ≈ 35 –45 Hz. All boards failed simultaneously, indicating a driver-level or SPI timing incompatibility rather than a single faulty board. The root cause is undiagnosed; we reverted to 110 kbps / 5000 μ s. **[TODO: If diagnosed before submission, replace with fix description.]**

Spatial error map lag. The Tier-2 correction $c_i(\hat{\mathbf{p}}_{t-1})$ uses the previous sweep’s EKF position (one-step lag, ≈ 100 ms). At walking speed this corresponds to ≈ 10 cm of motion — negligible for the IDW correction accuracy. At high speed or during abrupt turns, the correction lags by one sweep; the linear fade at grid boundaries limits the impact.

XII. CONCLUSION

We presented **DUNE** (Dual-tag UWB Navigation Engine), an ultra-wideband localization system delivering full 6-DOF pose estimation for a 13-DOF rocker-bogie mobile manipulator operating in unstructured terrain. DUNE addresses the fundamental limitations of all prior open-source Decawave DW1000 implementations through three focused contributions.

Dual-tag 6-DOF pose estimation. Two UWB tags at a 50 cm baseline on the rover chassis provide geometric yaw from their world-frame position difference. Combined with five-anchor 3D ranging (one elevated anchor making z observable) and BNO085 IMU fusion for roll and pitch, the system

outputs complete $(x, y, z, \phi, \theta, \psi)$ at ≈ 10 Hz—the first UWB implementation to provide full 6-DOF pose without motion capture.

Two-tier in-situ calibration. Tier 1 antenna-delay binary search eliminates the 30–50 cm bulk bias present on uncalibrated boards, requiring only a tape measure and taking under two minutes per board. Tier 2 spatial error mapping corrects residual position-dependent multipath. We document and remove a double-correction failure in naive prior approaches in which the same bias b_0 was both pushed to NVS and subtracted in software simultaneously, producing a net over-correction of $2b_0$ that inverted the bias direction entirely.

Same-frame NLOS detection and weighted multilateration. Reading first-path and receive power from the same `RANGE_REPORT` frame on the tag side yields a physically valid per-anchor NLOS gap signal, correcting an error in all existing libraries where the gap was computed across two different radio links. Continuous weighting in the Levenberg–Marquardt solver, combined with an independent MAD gate for gross outliers, delivers robust positioning under partial obstructions.

[TODO: Insert quantitative summary once hardware experiments complete: DUNE achieves X cm 2D RMSE and Y deg heading RMSE at Z Hz on dynamic rover trajectories over sandy terrain, compared to W cm for an uncalibrated single-tag baseline.]

Future work. Immediate next steps include extending the EKF from 2D to a 9-state 3D formulation once the five-anchor geometry is field-validated, tighter IMU–UWB coupling directly on raw ranges, and coordinated TDMA slot assignment for deterministic multi-tag operation. Longer-term, migration to the Qorvo DW3000 (pin-compatible, production-current) will extend the system lifetime beyond the DW1000 end-of-life.

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APPENDIX

Let T_{PS} , T_{PR} , T_{RS} denote the tag’s POLL send, POLL_ACK receive, and RANGE send timestamps (in DW1000 ticks, corrected by antenna delay). Let T_{PRA} , T_{AS} , T_{AR} denote the anchor’s POLL receive, POLL_ACK send, and RANGE receive timestamps. Define the four intervals:

$$\begin{aligned} R_1 &= T_{PR} - T_{PS}, & P_1 &= T_{AS} - T_{PRA}, \\ R_2 &= T_{AR} - T_{AS}, & P_2 &= T_{RS} - T_{PR}. \end{aligned}$$

Let δ_t and δ_a be the clock-frequency offsets of tag and anchor relative to the ideal frequency. The round-trip intervals satisfy $R_1 = (1 + \delta_t)[\tau + \tau + P_1/(1 + \delta_a)]$ (to first order). **[TODO: Write out the full first-order clock error cancellation. Show that Eq. (12) produces ToF accurate to $O(\delta^2\tau)$. Cite Neirynek 2016 or Decawave APS013 for the derivation.]**

The reported distance is $d = c \cdot \hat{\tau}$, where $c = 299,792,458$ m/s.

The residual for anchor i at estimated position $\mathbf{p}$ is $r_i(\mathbf{p}) = \|\mathbf{p} - \mathbf{a}_i\|_2 - \rho_i$. The Jacobian row for anchor i is

$$\mathbf{J}_i = \frac{(\mathbf{p} - \mathbf{a}_i)^\top}{\|\mathbf{p} - \mathbf{a}_i\|_2} \in \mathbb{R}^{1 \times d}. \quad (27)$$

The LM update at iteration k is

$$\Delta \mathbf{p} = -(\mathbf{J}^\top \mathbf{W} \mathbf{J} + \lambda \text{diag}(\mathbf{J}^\top \mathbf{W} \mathbf{J} + 10^{-9} \mathbf{I}))^{-1} \mathbf{J}^\top \mathbf{W} \mathbf{r}, \quad (28)$$

where $\mathbf{W} = \text{diag}(w_i)$ and λ is the Marquardt damping parameter. λ is divided by 3 on a successful step and multiplied by 3 on failure, bounded in $[10^{-9}, 10^9]$. Convergence is declared when $\|\Delta \mathbf{p}\| < 10^{-6}$ or after 30 iterations.

The output covariance is the GLS approximation: $\hat{\Sigma} = \hat{\sigma}^2(\mathbf{J}^\top \mathbf{W} \mathbf{J})^{-1}$, where $\hat{\sigma}^2 = \sigma_{\text{range}}^2 = (0.10 \text{ m})^2$ is the assumed range noise. This covariance is used as the EKF measurement noise matrix $\mathbf{R}$.